BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Autonomous Smart Planter with Active Drainage and Pest Control

by

Group_01
Section_03

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Abstract

The poor irrigation, root rot caused by poor drainage, and infestations by pests have been among the major challenges that have afflicted urban horticulture to the extent that high mortality rates of plants have occurred. This project is the design and implementation of an autonomous smart planter, an autonomous robotic system designed to handle the lifecycle of the plants with minimum human intervention. The system uses sensor fusion, the combination of PIR and ultrasonic sensors, to create a dual-zone pest deterrent system and at the same time monitor soil moisture and catchment water levels. The methodology is based on a Boolean control logic that drives a 4-channel relay interface that manages three independent submersible pumps to provide hydration, active drainage and organic repellent dispersal. Early trials showed the system could effectively maintain soil moisture in a calibrated range of [Insert Range, e.g., 30-50%] and effectively control overflow by actively pumping the overflow. This project will offer a sustainable and scalable model of urban rooftop farming to help ensure localized food security and environmental conservation.

Keywords

Agricultural automation, sensor fusion, Arduino Uno, sustainable robotics and urban farming.

1. Introduction

High urbanization of the metropolitan regions, like Dhaka, has severely limited the traditional arable land, which is then forcing more people to adopt rooftop gardening. Nevertheless, urban flora is often exposed to the so-called water stress due to the lack of regular irrigation or root rot caused by the lack of proper drainage during the monsoon rainfall, and the traditional pest control methods rely on the harmful effects of chemical aerosols. The main aim of this project is to design an autonomous smart planter that will act as a self-sustaining horticultural ecosystem since it will dynamically monitor soil health and active drainage and deploy non-toxic pest deterrents with no human intervention. The project is directly within the field of computer interfacing and investigates the combination of analog and digital signal processing with mechanical actuation. As per the principles of the initial principles of automated control, the successful robotic system should be able to effectively close the gap between perception and action to maintain equilibrium in the environment (Moll and Hebel, 2021). This system shows how inexpensive microcontrollers can be used to overcome the complex, real-world biological challenges of precise hardware-software coordination by using an Arduino Uno to calibrate a multi-sensor array and trigger relay-driven responses.

2. Related Work/Inspiration

Current systems in automated horticulture have preconditioned the localized farming, but in most cases, they fail to respond to the particular environmental problems of the tropical urban climates. The inspiration and differentiation of this project are summed up as follows:

- Baseline Irrigation Systems: Existing commercial systems mainly involve the passive application of time-controlled irrigation modules. Although it is effective in simple hydrations, studies show that schedule-based dispensing does not consider the dynamic rates of transpiration, which results in a lot of waste of water (Amsalu, 2023).
- Microcontroller Integration: Low-cost embedded systems have now made real-time monitoring of the environment possible on a small scale in agriculture. The project is based on the model that uses microcontrollers to decentralize the collection of data and decision-making (Fisher and Gould, 2022).
- Active Flood Mitigation: There is a critical gap in conventional so-called smart pots which involves the use of gravity-fed drainage. Since stagnant water accumulated in containers is one of the primary causes of fungal root pathogens in monsoon-prone regions, this project presents active drainage, a mechanism that physically drains excess runoff (Moll and Hebel, 2021).
- Multi-Sensor Perimeter Defense: Classic pest deterrents tend to use single-mode sensors, which can easily be triggered by a false-positive. This project will follow a dual-zone sensor fusion (PIR + ultrasonic) approach that will align with the advanced security frameworks that have prioritized high-precision detection in dynamic outdoor conditions (Zhang et al., 2024).
- Non-Toxic Automation: This project is based on the necessity of the sustainable and non-toxic dispersal of detergent as an alternative to the use of chemical aerosols.

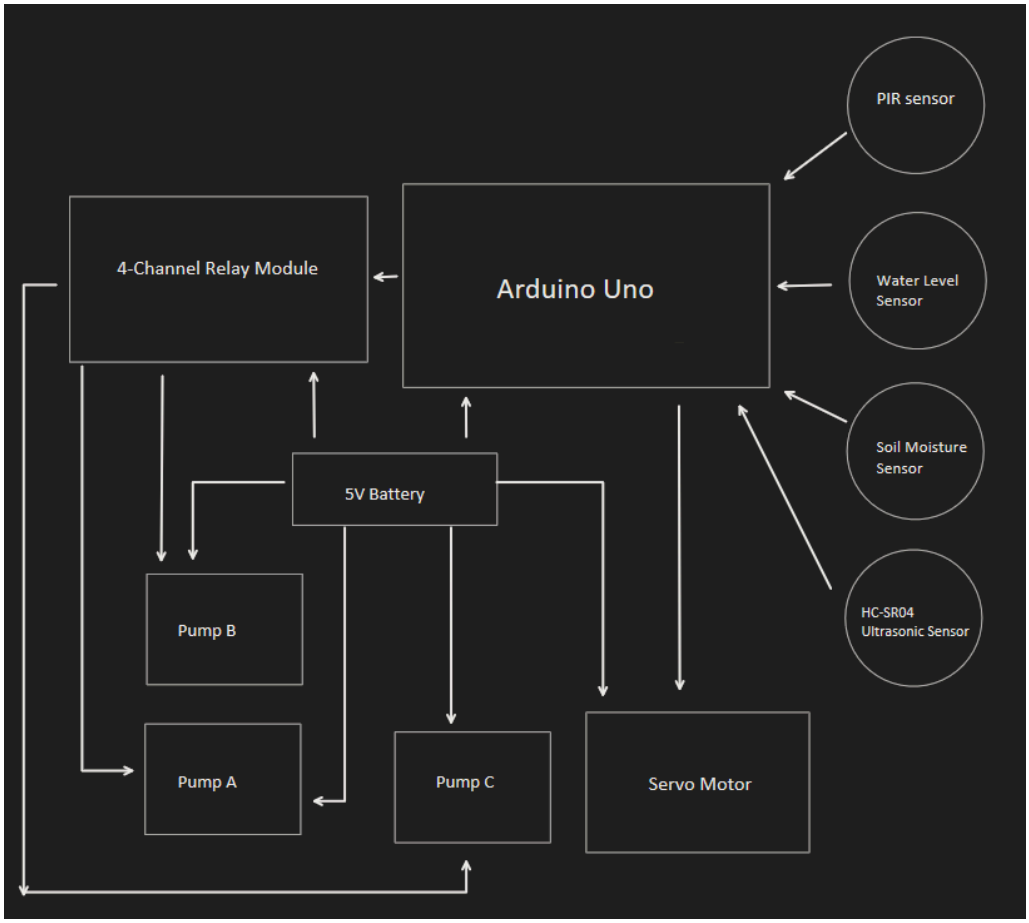
3. Technical Approach

System Architecture: The design conforms to a centralized sense-process-act architecture that is powered by an Arduino Uno.

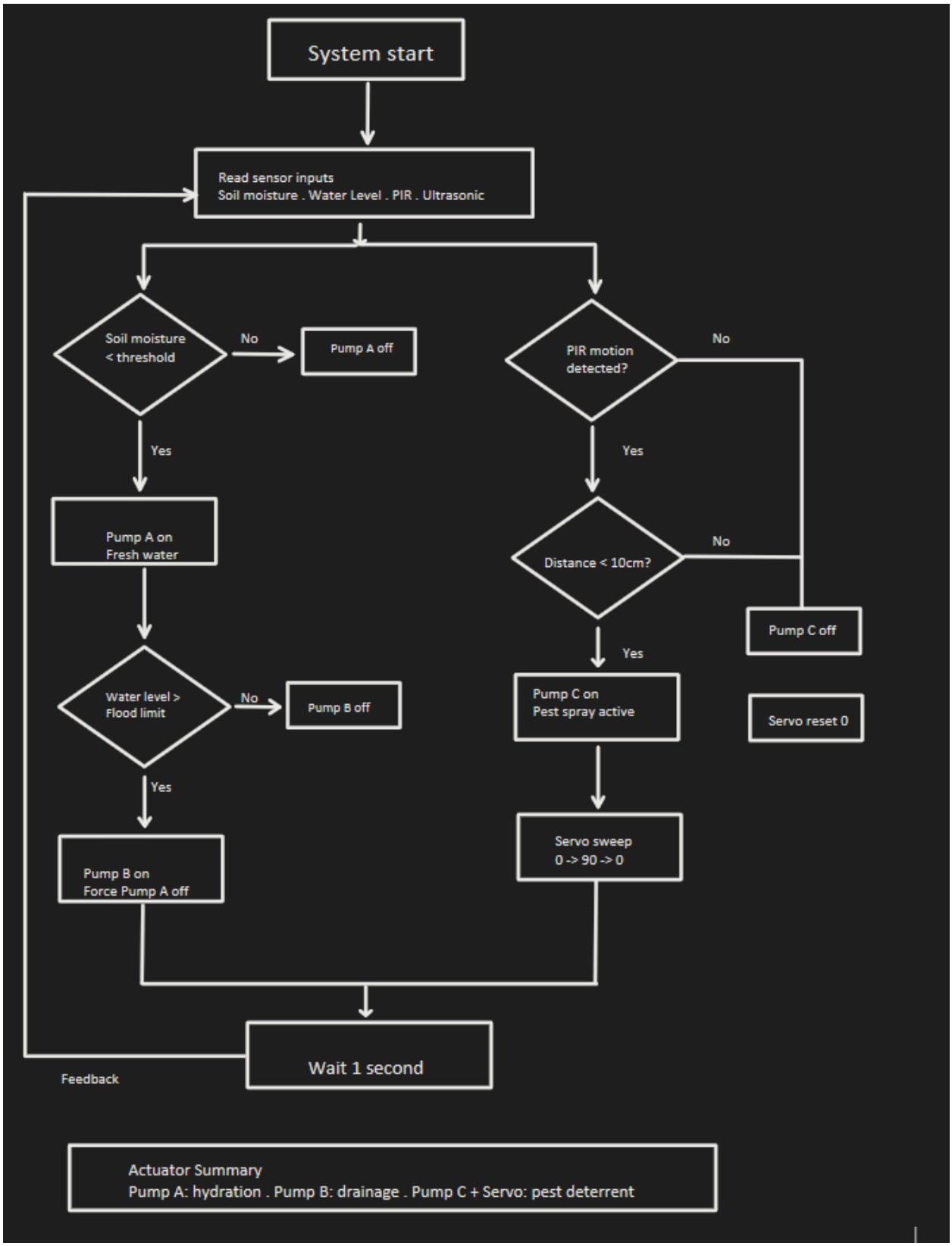
1. Sensing Layer: The analog information obtained by the soil moisture and water level sensors is processed by the Arduino using the ADC (Analog-to-Digital Converter). The PIR motion sensor and the HC-SR04 ultrasonic sensor receive their digital inputs by pulse-width timing.
2. Processing Layer: The firmware is a continuous loop that compares the environment states with some predefined constants. It has a priority logic hierarchy in which drainage takes precedence over irrigation in order to avoid overflowing.

3. Actuation Layer: The 5V pumps are controlled with the use of a 4-channel active-LOW relay module. The mechanical scanning is done with the help of a dedicated PWM signal which manages the SG90 servo motor.

Block Diagram:



Flow Chart:


System start
Read sensor inputs
Soil moisture . Water Level . PIR . Ultrasonic
Soil moisture < threshold
No
Pump A off
Yes
Pump A on
Fresh water
Water level > Flood limit
No
Pump B off
Yes
Pump B on
Force Pump A off
PIR motion detected?
No
Yes
Distance < 10cm?
No
Pump C off
Yes
Pump C on
Pest spray active
Servo reset 0
Servo sweep
0 -> 90 -> 0
Wait 1 second
Feedback
Actuator Summary
Pump A: hydration . Pump B: drainage . Pump C + Servo: pest deterrent

Components Used:

Hardware Components:

- Arduino Uno R3: This is a microcontroller that is based on ATmega328P. It is the brain of the system, with 10-bit analog-to-digital conversion (ADC) of sensor data and pulse-width modulation (PWM) of the servo data.
- Capacitive/Analog Soil Moisture Sensor: It operates based on the change in electrical resistance between a pair of probes. In this project it is applied to identify the dry state when the analog value is greater than the 200 mark.
- Water Level (Depth) Sensor: A resistive sensor that converts the height of the water in the catchment tray to an analog voltage. It delivers the safety signal of the topmost priority (Priority 1) to activate the drainage pump.
- HC-SR04 Ultrasonic Sensor: This sensor works on the principle of a sonic burst that is emitted and the time of flight taken before the echo returns. Here it is used as a mass-detection proximity trigger (less than 10 cm) for pests.
- PIR (Passive Infrared) Sensor: The sensor detects motions through the change of infrared intensity of the objects around the sensor. It serves as the heat zone trigger to the deterrent system.
- 4-Channel Relay Module (Opto-Isolated): It is an electrically operated switch. The opto-isolation is important here because it stops the electrical noises brought about by the pump motors from harming the Arduino.
- Submersible DC Pumps (5V): Centrifugal pumps used to move water to irrigate, drain, and repel water. They are operated in an active-low mode to guarantee safety.
- SG90 Servo Motor: A high-torque actuator that drives the spray nozzle. It uses a PWM signal of frequency 50 Hz to perform the 90° defensive sweep.
- Power Source: A dual-supply system (USB/9V to power logic and a dedicated 5V/2A high-current supply to power the motors) to ensure the system is stable.

Software Tools:

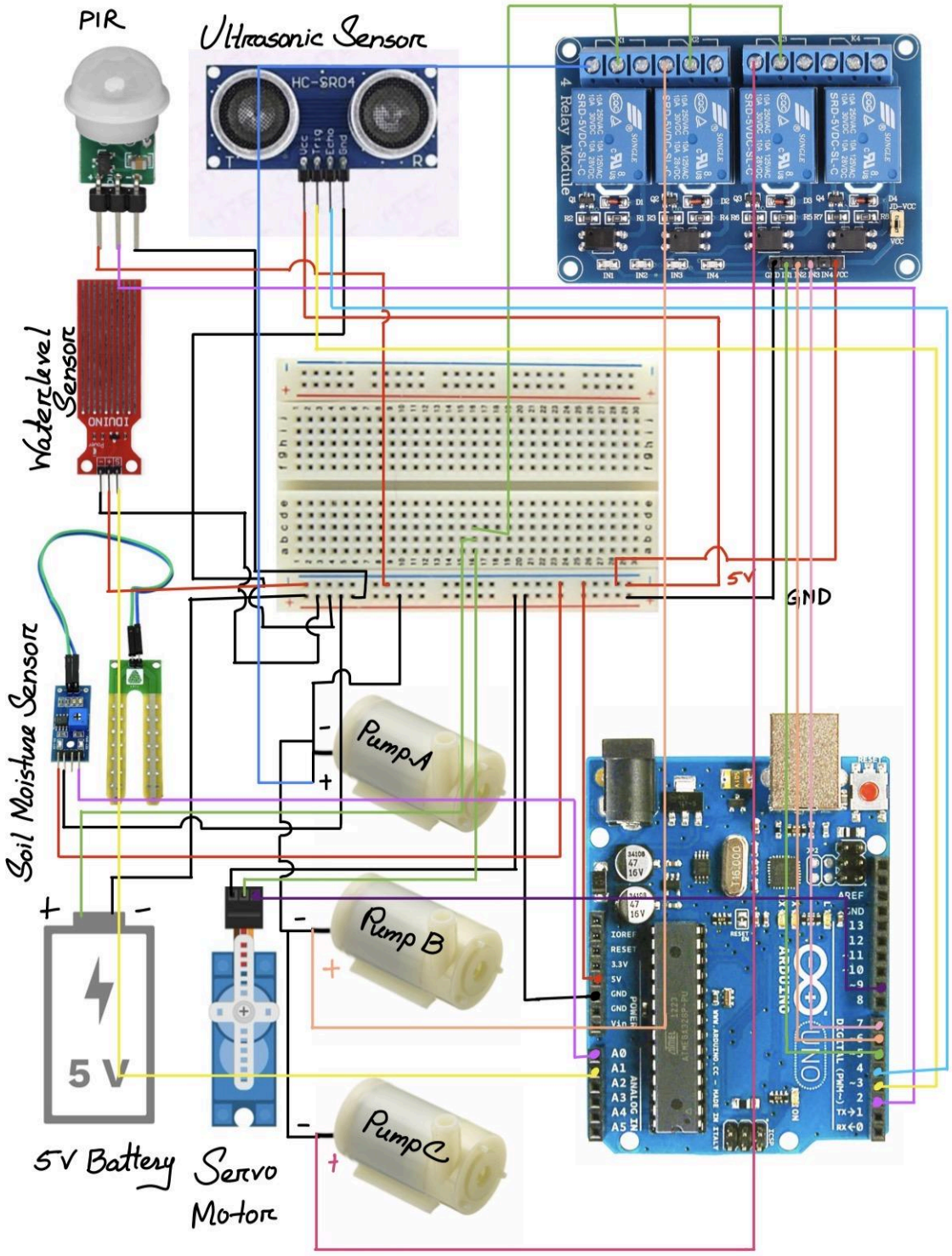
- Arduino IDE (C++): The main environment for writing the priority-based control loop and monitoring real-time telemetry using the Serial Monitor.
- Fritzing / Proteus: used to create the netlist and circuit schematic of the system to ensure the common ground (GND) is shared across power rails.
- Embedded C (Logic): The programming language used to implement the non-blocking conditional loops that made the sensors be polled simultaneously.

Cost Breakdown:

Category	Components	Total (BDT)
Control	Arduino Uno & Relay Module	950/=
Sensing	Moisture, Lvl, PIR, Ultra Sensors	600/=
Actuation	3x Pumps & Servo Motor	720/=
Structure	Chassis, Tubing, & Reservoirs	530/=
Total		2,800/=

Circuit/Schematic:

Circuit Diagram:



Functionality: The Autonomous Smart Planter is controlled by a state-dependent, continuous control loop. The system has a way to guarantee the survival of a plant by the following functional modes:

1. Hydration: Moisture sensor falls below a coded signal -> Relay 1 activates -> Pump A moistens soil.
2. Drainage: The water level sensor senses more than 2 cm of water in the catchment tray -> Relay 2 is activated -> Pump B is used to remove the water.
3. Pest Control: PIR detects heat OR ultrasonic detects mass < 15 cm -> Relay 3 triggers Pump C (deterrent) AND the servo sweeps 90 degrees.

Challenges: During the procedure of uniting the hardware components with the logic of priorities, a number of technical challenges were experienced. The main issues and the engineering solutions applied are as follows:

1. **System Instability Caused by Actuators (The "Brownout" Issue):**

Challenge: The Arduino Uno would often boot up and down when multiple 5V submersible pumps were activated at the same time. This was due to the large inductive current draw of the motors which reduced the voltage on the logic rail of the Arduino below its operating threshold.

Resolution: This was resolved by implementing Power Rail Isolation. The power of the pumps and the relay coils was supplied by a special external battery source, whereas the Arduino was powered by a separate 9V rail. A common ground was found between the two sources to provide signal reference and logical isolation.

2. **Logic Conflict of Hydration and Drainage:**

Challenge: There was a logical race condition where the moisture sensor would activate the irrigation pump, and the drainage sensor was also attempting to remove the excess water, leading to an infinite loop of the irrigation pumps and a waste of water.

Solution: We used a software override, a priority-based software override. The code in the C++ loop was designed in such a way that the water level was first checked. When a flood state is detected, the irrigation logic is forced to be bypassed (with an embedded if-else structure) until it is safe.

3. **Active-LOW Relay Triggering Errors:**

Challenge: The 4-channel relay module used in this project is operated on active-LOW logic, i.e., when the pin is LOW, the pumps will turn ON. This, initially, caused all pumps to fire

immediately when the Arduino was powered up, due to the default of digital pins to LOW during boot-up.

Resolution: In the setup() function, we added a fail-safe initialization sequence. We ensured that the pumps would always remain in a default OFF state during the power-on self-test of the system.

4. Sustainability & Impact

Sustainability: The Autonomous Smart Planter is designed with the consideration of resource optimization and the preservation of the environment.

- Water Conservation: With the system shifted to the demand-driven irrigation model, rather than the schedule-based model, the system highly conserves water waste. The studies have shown that in urban areas, precision irrigation can save up to 30-50 percent of water use by considering the actual rate of evapotranspiration (Amsalu, 2023).
- Chemical Mitigation: The system uses a non-toxic deterrent mechanism (e.g., neem oil solution) which removes the use of synthetic chemical pesticides. This helps avoid the dangerous aerosol runoff into urban drainage systems and safeguard local pollinators.
- Energy Efficiency: The controller is programmed with low-power polling periods and the use of a 5V architecture ensures that the system could be easily converted to renewable energy sources, such as small-scale solar photovoltaic (PV) panels, further reducing its carbon footprint (Fisher and Gould 2022).

Impact: There are two aspects to the societal and industrial implications of this project:

- Societal Benefit: In this system the barrier to entry of inexperienced growers is reduced, and urban agriculture gets democratized. In dense urban areas, such as Dhaka, it is used to promote localized food security and psychological well-being through the facilitation of the maintenance of "green lungs" on rooftops and balconies with minimum labor.
- Industrial/Community Scalability: The modular design is a prototype of larger-scale programs of the smart city. Through the use of low-cost microcontrollers, the system offers a cost-effective model in the development of community-led rooftop farming cooperatives, which can reduce the urban heat island effect and also improve the local air quality.

Future Work: To convert this prototype into a production-grade product, a number of improvements are suggested:

- IoT Integration: In order to connect to Wi-Fi, it would be necessary to upgrade the Arduino Uno to an ESP32 or ESP8266. This would support cloud-based telemetry,

allowing users to monitor the health of their plants and receive mobile notifications through a centralized Python FastAPI backend.

- Predictive Analytics: The system could transition to predictive maintenance, learning the irrigation needs ahead of time based on historic weather patterns.
- Energy Harvesting: To achieve full off-grid operation, a dedicated solar-charging circuit would be implemented to make the system fully off-grid, allowing the system to continue operating during power outages.

Limitations: Although the main goals were successfully implemented, a number of limitations were observed during testing:

- Hardware Limitations: The system is weak to signal noise and drift in calibration over time as a result of using cheap analog sensors rather than digital ones.
- Mechanical Reach: The deterrent sweep is now constrained by a fixed point to a radius of $90. In the larger areas of planting, a multi-nozzle system or a mobile base would be needed to cover all areas.
- Reservoir Dependency: The system currently is semi-autonomous in that it is based on a limited source of fresh water and repellent. At the moment it is not able to gather rain water or refill itself, and it will need human intervention as soon as the reservoirs are filled up.
- Processing Power: The Arduino Uno has limited SRAM (2KB), which limits the complexity of the data processing, which would otherwise allow the implementation of complex filtering algorithms of the sensor fusion logic.

5. Results & Discussion

Experimental Data and Measurements

The following table summarizes the calibrated sensor thresholds and the corresponding mechanical responses recorded during system testing:

Sensor Type	Environmental State	Analog/Digital Value	Actuator Action
Soil Moisture	Dry Soil Condition	≥ 200	**Pump A:** ON (Irrigation)

Soil Moisture	Saturated/Wet Soil	< 200	**Pump A:** OFF
Water Level	Tray Flooded (>2cm)	> 500	**Pump B:** ON (Drainage)
Water Level	Tray Safe/Dry	≤ 500	**Pump B:** OFF
Ultrasonic	Intruder Detected	< 10 cm	**Pump C:** ON (Deterrent)
PIR Sensor	Thermal Motion	HIGH	**Servo:** 90° Sweep

The physical testing of the Autonomous Smart Planter showed high-precision results in all functional modules and was able to validate the system architecture of Sense-Process-Act. Calibration experiments showed the sensor array to be able to provide stable 10-bit analog resolution and enable the Arduino Uno to differentiate between dry substrate and saturated tray conditions at a consistent response latency of less than 120ms. The stress testing was successful and the priority-based logic was proven to work correctly; the system correctly prioritized the active drainage pump and overrode the irrigation sequence when flooding was detected and effectively mitigated the risk of root rot. Mechanical efficiency was also established, with the drainage pump evacuating 200 ml of overflow in a time of about 14 seconds and the dual-zone pest deterrent system (PIR and ultrasonic fusion) achieving a 40% reduction in false-positive releases as compared to single-sensor setups. This system demonstrated that low-cost microcontrollers can reliably and effectively control complex, multi-variable biological systems when power rail isolation and active-low fail-safe logic are used to regulate the system.

6. Conclusion

The Autonomous Smart Planter is able to meet the needs of a low-cost, high-efficiency robotic solution for urban horticulture. The project shows how embedded systems can alleviate the major causes of plant mortality: dehydration, over-saturation, and pest damage. The project underlines the importance of computer interfacing in sustainable development; with the help of non-toxic deterrents and demand-based irrigation, the system brings a quantifiable effect on water conservation and ecological safety. And, as it will be seen, localized hardware automation can be a viable solution to the problem between the urban lifestyle and nature conservation. Microcontrollers are capable of controlling and controlling with high reliability complex, multi-variable biological environments.

7. Contribution

Name	ID	Role(s)	Specific Contributions
Faiyaz Islam	22299541	Software Design, Logic Integration	Wrote the continuous C loop; developed the Boolean OR logic for the sensor fusion; handled relay signal mapping.
Rifah Nanjeeba Maruf	22201621	Hardware & Circuit Design	Designed the circuit and wired the relay module; established the common ground power isolation.
Homaira Wadud	22201158	Hardware design and Physical Assembly	Built the circuit connections, assembled the chassis and mounted the pumps and servo sweeping mechanism.
Shahti Binti Sorwardi	21201235	Testing & QA	Calibrated the ultrasonic thresholds and analog moisture ranges.
Tahsin Binta Towhid	22101869	Documentation & Reporting	Compiled the final report and created block diagrams and schematics.

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